

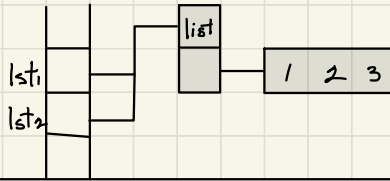
# Memory Maps

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Online Lecture  
S. Romero Cruz  
CS-UY 1134  
01/26/26 

lst1 = [1, 2, 3]

lst2 = lst1



## Shallow Copy

A new list object is created but the contents of both list are shared.

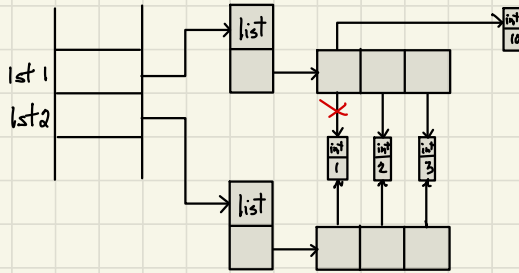
`import copy`

`#copy.copy(...)`

lst1 = [1, 2, 3]

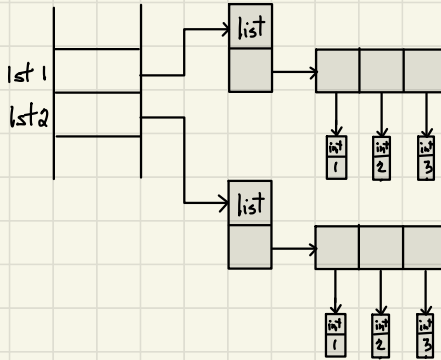
lst2 = copy.copy(lst1)

lst1[0] = 10



lst1 = [1, 2, 3]

lst2 = copy.deepcopy(lst1)



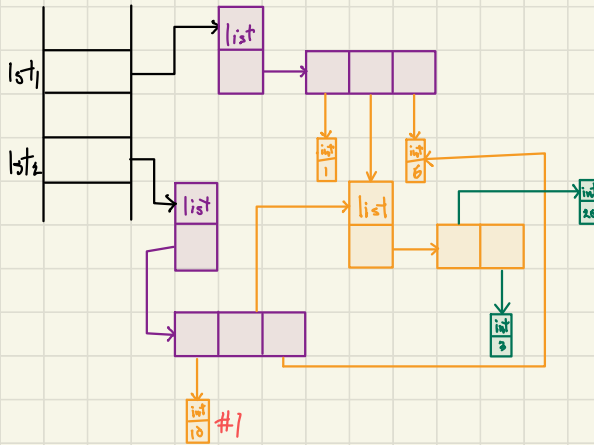
## Shallow Copies w/ Nested Lists

lst1 = [1, [2, 3], 6]

lst2 = copy.copy(lst1)

lst2[0] = 10 #1

lst2[1][0] = 20 #2



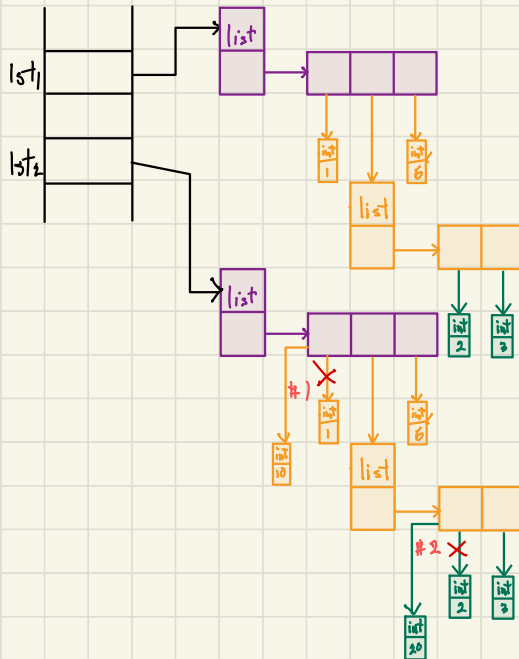
## Deep Copies w/ Nested Lists

lst1 = [1, [2, 3], 6]

lst2 = copy.deepcopy(lst1)

lst2[0] = 10 #1

lst2[1][0] = 20 #2



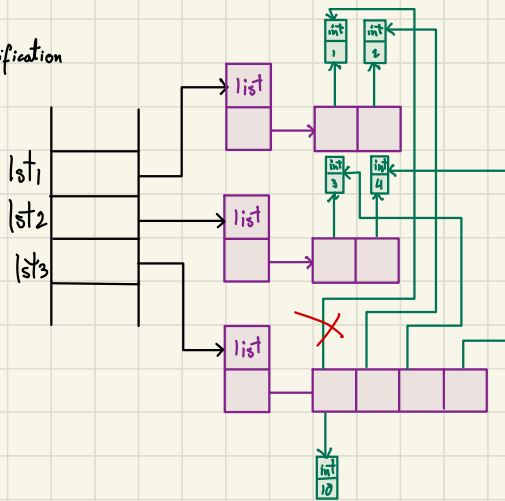
## List Addition & Modification

$lst_1 = [1, 2]$

$lst_2 = [3, 4]$

$lst_3 = lst_1 + lst_2$

$lst_3[0] = 10$



class Counter:

```
def __init__(self):  
    self.value = 0
```

```
def inc(self):  
    self.value += 1
```

```
def __repr__(self):  
    return str(self.value)
```